QP CODE: 24900144

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course-MG1DSCHPY100 - BASICS OF PHYSIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks :50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap) and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability

Part A

Multiple Choice Questions
Answer all questions
Each question carries 1 mark

1. Digestion of waste
2. To transfer information from one cell to another
3. Globulin
4. Poikilocytes
5. Scapula
6. Osteon

Part B

Short Answer Questions
Answer two questions
Each question carries 2 marks

7. ***½ mark each for each points***

   i) Framework for the body
   ii) Support and protect organs
   iii) Serves as attachment points for muscles
   iv) Enable flexibility and various ranges of motion

8. ***Any four points – ½ mark each***

   i) Antibody formation
   ii) Phagocytosis
   iii) Form of fibroblast
   iv) Manufacture of trephines
   v) Secretion of heparin

9. ***½ mark each for each points***

   i) Osteocytes are primary cell of mature bone and the most common type of bone cell.

ii) Osteocyte is located in a small cavity in the bone tissue called a lacuna.

iii)Maintain the mineral concentration of the matrix via the secretion of enzymes.

iv) Communicate with each other and receive nutrients via long cytoplasmic processes that extend through canaliculi

10. ***½ mark each for each points***

i) Cardiac muscle forms the musculature of the heart.

ii) These muscles are short, cylindrical, striated, involuntary.

iii) Each fibre is surrounded by the sarcolemma which encloses the sarcoplasm with longitudinal myofibrils, a single nucleus, several mitochondria and well developed sarcoplasmic reticulum.

iv) Cardiac muscles are involuntary, they can go on contracting and relaxing throughout life and are immune to fatigue.

Part C

Short Essay Type Questions

Answer four questions

Each question carries 5 marks

11. ***1 mark each for any five points***

- Structure and support
- Protecting vital organs (e.g., skull protects the brain)
- Facilitates movement by serving as attachment points for muscles
- Store minerals (like calcium)
- Produce blood cells in the bone marrow,
- Contributing to overall health and homeostasis.

12. ***1 mark each for each points***

- Tight junctions (prevent leakage)
- Adherens junctions (mechanical strength)
- Desmosomes (resist tension)
- Gap junctions (cell communication)
- Hemidesmosomes (attach cells to the matrix)

13. i) Intracellular fluid – **1 mark**

ii) Extracellular fluid – Plasma, interstitial fluid, transcellular fluid, mesenchymal fluid – **4 marks**

14. ***½ mark each for each points***

i) Essential for blood clotting

ii) Maintain colloidal osmotic pressure

iii) Maintain viscosity and blood pressure

iv) Maintain acid-base balance by buffering action

v) Act as protein store- the plasma proteins serve as store house of proteins.

vi) Trasport of CO_2- proteins helps in CO_2 carriage by forming carbamino Hb.

vii) Rouleaux formation- tendency of RBCs to adhere to one another

viii) Antibodies like gama-globulin is present in plasma which act as defense against infection

ix) Help transport of certain substance like transferring and ceruloplasmin

x) Regulate the distribution of fluid between blood and tissue

15. Composition of bone – matrix and bone salt **- 2 marks**

Functions of bone – **3 marks**

1. **Protective function**: Protects soft tissues and vital organs of the body.
2. **Mechanical function**: Supports the body and brings out various movements of the body by their attachment to the muscles and tendons.
3. **Metabolic function**: Plays an important role in the metabolism homeostasis of calcium and phosphate in the body.
4. **Hemopoietic function**: Red bone marrow in the bones is the site of production of blood cells.

16. ***1 mark each for each points***

7 Cervical vertebrae

12 Thoracic vertebrae

5 Lumbar vertebrae

1 Sacral vertebrae

1 Coccyx

Part D

Long Essay Questions
Answer two questions
Each question carries 10 marks

17. Structure of nucleus – nucleoplasm, nuclear envelope, nucleolus, chromatin 8 marks – **(8 marks – 2 marks for each points)**

    Labelled diagram – **2 marks**

18. Oedema is defined as the swelling caused by excessive accumulation of fluid in the tissues. – **1mark**

    Types – intracellular and extracellular oedema – **2 marks**

    Causes – **7 marks**

    i) Increased capillary permeability
    ii) Increase in the capillary pressure
    iii) Blockage of lymphatic nodes or vessels, as a result of inflammation of the node
    iv) Loss of the plasma proteins due to malnutrition or renal damage
    v) Renal diseases
    vi) Unfamiliar exercise
    vii) Ingestion of large quantity of salt

19. Composition of bone – matrix and bone salt – **2 marks**

    Structure **– 3 marks**

    Classification – long bone, short bone, flat bone, irregular bone, sesamoid bone – **5 marks (1 mark for each type of bone)**

20. Structure and function of

    Smooth muscle with diagram - **3 marks**

    Skeletal muscle with diagram - **4 marks**

    Cardiac muscle with diagram - **3 marks**